

# An Entangled Affine Arboreal Image for $z \mapsto z^2$ at Basepoint 2

HCS-C374 / HEN-O358

4 September 2026

## Abstract

For the  $n$ th preimages of 2 under  $z \mapsto z^2$ , we determine the splitting field and Galois action uniformly for every  $n \geq 3$ . The pure radical and cyclotomic fields meet in exactly  $\mathbb{Q}(\sqrt{2})$ , and this shared quadratic field cuts the full affine group by the character equation  $(-1)^b = (2/a)$ . The resulting image has index two in  $\text{AGL}_1(\mathbb{Z}/2^n\mathbb{Z})$ , and adjacent-level restriction is surjective with kernel four. We further classify the complete fixed-root spectrum and prove that the odd primes for which  $x^{2^n} = 2$  has a root have density  $7/24 + 1/(3 \cdot 4^{n-1})$ , tending to  $7/24$ ; exactly one and four roots never occur.

**Keywords:** arboreal Galois representations; Kummer extensions; Frobenius fixed points; Chebotarev density; affine groups; iterated preimages

## 中文摘要

本文考察平方映射在基点二之上的逐层原像域。第  $n$  层的根可用  $2^n$  次单位根编号；纯根式域与分圆域恰好共享  $\mathbb{Q}(\sqrt{2})$ ，这一个二次交叠迫使仿射参数满足模八的二次特征等式。由此得到指数二的精确Galois像，并证明相邻层限制满射且核的阶为四。进一步把Frobenius不动根化为一个二进线性同余。完整计数只允许零、二以及从八起的二次幂根数，特别排除一根与四根情形。结合Chebotarev定理，所得素数条件自然密度为  $7/24 + 1/(3 \cdot 4^{n-1})$ ，并单调趋于  $7/24$ 。  
关键词：算术动力学；Frobenius不动点；Chebotarev密度；二次特征；根数分布；素数密度

## 1 Preimage fields and the shared quadratic field

Let  $m = 2^n$ , choose compatible elements

$$\alpha_n = 2^{1/m}, \quad \zeta_m = e^{2\pi i/m},$$

and put

$$R_n = \{\zeta_m^j \alpha_n : j \in \mathbb{Z}/m\mathbb{Z}\}, \quad K_n = \mathbb{Q}(R_n) = \mathbb{Q}(\alpha_n, \zeta_m).$$

These are the level- $n$  vertices and splitting fields of the preimage tree of 2 for  $z \mapsto z^2$ ; see Jones [1] for the general arboreal framework. The present monomial tower is solvable, but it is not the generic full-affine Kummer tower:  $\alpha_n^{m/2} = \sqrt{2}$  already lies in the cyclotomic field.

**Lemma 1** (Cyclotomic nonsquare). *For every  $n \geq 3$ ,  $\sqrt{2}$  lies in  $C_n := \mathbb{Q}(\zeta_m)$  but is not a square in  $C_n$ .*

*Proof.* Using the standard cyclotomic description [3], because  $\zeta_8 = \zeta_m^{2^{n-3}}$ ,  $\sqrt{2} = \zeta_8 + \zeta_8^{-1} \in C_n$ . If  $y^2 = \sqrt{2}$  in  $C_n$ , then  $y = \pm 2^{1/4}$ , so  $\mathbb{Q}(2^{1/4})$  would be an intermediate field of the abelian Galois extension  $C_n/\mathbb{Q}$ . Every such intermediate extension is Galois. This is impossible:  $X^4 - 2$  is Eisenstein at 2, while the real field  $\mathbb{Q}(2^{1/4})$  does not contain its conjugate  $i2^{1/4}$ .  $\square$

**Theorem 1** (Radical–cyclotomic intersection). *For all  $n \geq 3$ ,*

$$\mathbb{Q}(\alpha_n) \cap \mathbb{Q}(\zeta_m) = \mathbb{Q}(\sqrt{2}), \quad [K_n : \mathbb{Q}] = 2^{2n-2}.$$

*Proof.* Over  $C_n$ ,  $\alpha_n$  satisfies  $X^{m/2} - \sqrt{2}$ . Capelli's binomial criterion [2] reduces irreducibility to excluding a square and, when  $4 \mid m/2$ , the exceptional shape  $\sqrt{2} = -4c^4$ . The first is Lemma 1. The second would also make  $\sqrt{2}$  a square, namely  $(2ic^2)^2$ , because  $i \in C_n$ . Hence  $[K_n : C_n] = m/2$ , and

$$[K_n : \mathbb{Q}] = (m/2)\varphi(m) = 2^{2n-2}.$$

Eisenstein gives  $[\mathbb{Q}(\alpha_n) : \mathbb{Q}] = m$ . The compositum formula now gives intersection degree two; the contained field  $\mathbb{Q}(\sqrt{2})$  is therefore the whole intersection.  $\square$

## 2 Exact affine image and compatible restrictions

For odd  $a$ , let

$$\chi_2(a) = \left(\frac{2}{a}\right) = (-1)^{(a^2-1)/8},$$

viewed modulo 8. Define

$$H_n = \{(a, b) \in (\mathbb{Z}/m\mathbb{Z})^\times \ltimes \mathbb{Z}/m\mathbb{Z} : (-1)^b = \chi_2(a)\},$$

where  $(a, b)$  sends  $j$  to  $aj + b$ .

**Theorem 2** (Exact image and restriction). *The action on  $R_n$  identifies  $\text{Gal}(K_n/\mathbb{Q})$  with  $H_n$ . This group has order  $2^{2n-2}$  and index two in  $\text{AGL}_1(\mathbb{Z}/m\mathbb{Z})$ . Under compatible choices  $\alpha_n = \alpha_{n+1}^2$  and  $\zeta_{2n} = \zeta_{2n+1}^2$ , restriction is*

$$H_{n+1} \longrightarrow H_n, \quad (a, b) \longmapsto (a, b) \pmod{2^n}.$$

*It is surjective and its kernel has order four.*

*Proof.* Every automorphism has unique parameters  $\sigma(\zeta_m) = \zeta_m^a$  and  $\sigma(\alpha_n) = \zeta_m^b \alpha_n$ , hence acts by  $j \mapsto aj + b$ . The two expressions for the shared square root give

$$\sigma(\sqrt{2}) = (-1)^b \sqrt{2} = (\zeta_8^a + \zeta_8^{-a}) = \chi_2(a) \sqrt{2}.$$

Thus the image lies in  $H_n$ . There are  $\varphi(m)$  choices of  $a$  and  $m/2$  choices of the forced parity of  $b$ , so  $|H_n| = 2^{2n-2} = [K_n : \mathbb{Q}]$ . The faithful inclusion is an equality.

Squaring the radical and root-of-unity formulas proves coordinate reduction. Each coordinate has two lifts. Reduction preserves  $a$  modulo 8 and the parity of  $b$ , so all four lifts satisfy the same character equation. This proves surjectivity and the kernel count.  $\square$

**Round 0 advance: field and compatible-image closure.** The intersection, every finite image, and every adjacent restriction map are proved for all levels  $n \geq 3$ ; no finite-level extrapolation occurs.

## 3 The complete fixed-root spectrum

For  $(a, b) \in H_n$ , fixed roots correspond to solutions of

$$(a-1)j \equiv -b \pmod{2^n}. \tag{1}$$

**Theorem 3** (Fixed-root law). *The exact numbers of elements of  $H_n$  having a prescribed number of fixed level- $n$  roots are*

| fixed roots         | number of elements                                       |
|---------------------|----------------------------------------------------------|
| 2                   | $2^{2n-4}$                                               |
| $2^k, 3 \leq k < n$ | $2^{2n-2k-1}$                                            |
| $2^n$               | 1                                                        |
| 0                   | $2^{2n-2} - 2^{2n-4} - \sum_{k=3}^{n-1} 2^{2n-2k-1} - 1$ |

*No element has exactly one or four fixed roots.*

*Proof.* Equation (1) has  $g = \gcd(a-1, 2^n)$  solutions exactly when  $g \mid b$ . Since  $a$  is odd,  $g \geq 2$ , excluding one root. If  $a \equiv 7 \pmod{8}$ , then  $\chi_2(a) = 1$ ,  $b$  is even, and  $v_2(a-1) = 1$ . The  $2^{n-3} \cdot 2^{n-1} = 2^{2n-4}$  such pairs have two fixed roots.

For  $3 \leq k < n$ , there are  $2^{n-k-1}$  residues with  $v_2(a-1) = k$  and  $2^{n-k}$  admissible  $b$  divisible by  $2^k$ , giving the second row. When  $a = 1$ , only  $b = 0$  fixes all roots. Finally,  $v_2(a-1) = 2$  means  $a \equiv 5 \pmod{8}$ , so  $\chi_2(a) = -1$  forces  $b$  odd; then  $4 \nmid b$ , excluding four fixed roots. The complement has none.  $\square$

## 4 Chebotarev root densities

All odd primes are unramified in  $K_n$ , since the polynomial discriminant has no odd prime divisor. Its reduction is separable, and  $x^{2^n} = 2$  has a root over  $\mathbb{F}_p$  exactly when arithmetic Frobenius fixes a member of  $R_n$ .

**Theorem 4** (Root-prime density). *For every  $n \geq 3$ , the odd primes at which  $x^{2^n} = 2$  has a root have natural density*

$$\delta_n = \frac{7}{24} + \frac{1}{3 \cdot 4^{n-1}}. \quad (2)$$

*The sequence decreases to  $7/24$ . Each nonzero root multiplicity in Theorem 3 separately has the density obtained by dividing its group count by  $2^{2n-2}$ .*

*Proof.* The fixed-point property and fixed-point count are conjugacy invariant. Chebotarev [4] therefore identifies the requested densities with the corresponding proportions in  $H_n$ . The positive fraction is

$$\frac{1}{4} + \sum_{k=3}^{n-1} 2^{1-2k} + 2^{-2n+2} = \frac{7}{24} + \frac{1}{3 \cdot 4^{n-1}}.$$

The limit and monotonicity are immediate. □

**Round 1 advance: fixed-root and density closure.** The complete Frobenius fixed-root spectrum, the impossible multiplicities 1 and 4, every multiplicity density, and the limit  $7/24$  are added as all-level theorems.

## 5 Conclusion

The basepoint 2 produces one persistent quadratic entanglement and no further loss: it cuts the affine image by exactly one character at every level. The resulting compatible group law turns a radical-field question into a complete fixed-root and prime-density theorem. The density converges to  $7/24$ , while the character obstruction explains the otherwise surprising absence of exactly four roots.

## References

- [1] R. Jones, *Galois representations from pre-image trees: an arboreal survey*, Publications mathématiques de Besançon (2013), 107–136, doi: <https://doi.org/10.5802/pmb.a-154>.
- [2] S. Lang, *Algebra*, third edition, Graduate Texts in Mathematics 211, Springer, 2002, doi: <https://doi.org/10.1007/978-1-4613-0041-0>.
- [3] L. C. Washington, *Introduction to Cyclotomic Fields*, second edition, Graduate Texts in Mathematics 83, Springer, 1997, doi: <https://doi.org/10.1007/978-1-4612-1934-7>.
- [4] J. Neukirch, *Algebraic Number Theory*, Springer, 1999, doi: <https://doi.org/10.1007/978-3-662-03983-0>.